

This unit is assessed by means of a written examination paper of 1 hour 30 minutes duration. The paper will consist of objective, short-answer and long-answer questions. Students may be required to apply their knowledge and understanding of physics to situations that they have not encountered before. The total number of marks available for this examination paper is 80. It contributes 40% to IAS and 20% to the IAL in Physics.

It is recommended that students have access to a scientific calculator for this paper.

Students will be provided with the formulae sheet shown in *Appendix 5*. Any other physics formulae that are required will be stated in the question paper.

The quality of written communication will be assessed in the context of this unit through questions which are labelled with an asterisk (*). Students should take particular care with spelling, punctuation and grammar, as well as the clarity of expression, on these questions.

6.3 The Sound of Music (MUS)

A study of music and recorded sound, focusing on the production of sound by musical instruments and the operation of a CD player:

- synthesised and 'live' sounds
- standing waves in string and wind instruments
- reading a CD by laser.

Waves and photons are used to model the behaviour of light.

There are opportunities for students to develop ICT skills and other skills relating to practical investigation and to communication.

Students should discuss environmental issues related to noise.

Students will be assessed on their ability to:		Suggested experiments
28	understand and use the terms <i>amplitude</i> , <i>frequency</i> , <i>period</i> , <i>speed</i> and <i>wavelength</i>	Wave machine or computer simulation of wave properties
29	identify the different regions of the electromagnetic spectrum and describe some of their applications	
30	use the wave equation $v = f\lambda$	
31	recall that a sound wave is a longitudinal wave which can be described in terms of the displacement of molecules	Demonstration using a loudspeaker Demonstration using waves on a long spring
32	use graphs to represent transverse and longitudinal waves, including <i>standing (stationary) waves</i>	
33	explain and use the concepts of wavefront, coherence, path difference, superposition and phase	Demonstration with ripple tank
34	recognise and use the relationship between phase difference and path difference	
35	explain what is meant by a <i>standing (stationary) wave</i> , investigate how such a wave is formed, and identify nodes and antinodes	Melde's experiment, sonometer